

Computer Organization and Assembly Language

Lecture 2

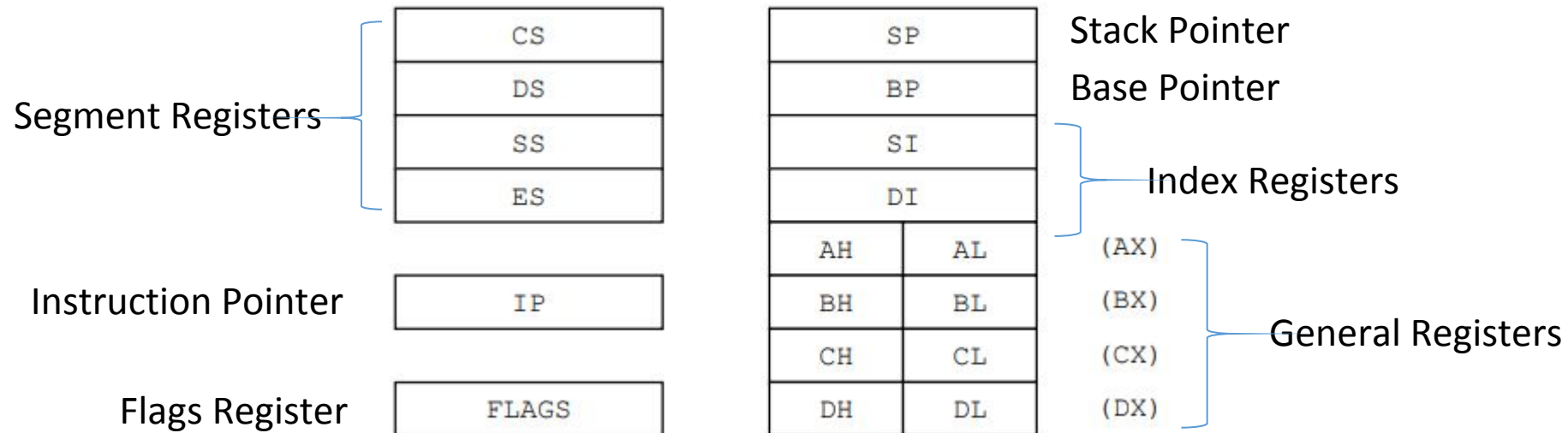
This lecture will cover

- INTEL IAPX88 ARCHITECTURE (1.5 BH)
 - Registers (1.2, 1.6 BH)
 - Buses and Memory
- First program in Assembly (1.7 BH)
 - Introduction to DosBox, NASM, AFD

INTEL IAPX88 ARCHITECTURE

IAPX88 REGISTER ARCHITECTURE

- 8088 is a 16bit processor with its accumulator and all registers of 16 bits
- Consists of 14 registers.



Registers

- Types of Registers

- Accumulator

- Central register in every processor
 - Traditionally all mathematical and logical operations are performed on the accumulator
 - n bit processor has n bit accumulator. For example a 16 bit processor has 16 nit accumulator.

- Pointer, Index, or Base Register

- Holds the address of data

- Flag Register

- This is a special register in every architecture called the flags register or the program status word.
 - The bits of the flags register work independently and individually, and combined its value is meaningless.
 - An example of a bit commonly present in the flags register is the carry flag
 - If a 16bit number is added to a 16bit accumulator, and the result is of 17 bits the 17th bit is placed in the carry bit of the flags register.

- Program Counter or Instruction Pointer

- The program counter holds the address of the next instruction to be executed

General Registers (AX, BX, CX, and DX)

- Used for arithmetic and data movement
- X in their names stand for extended meaning 16bit registers
 - For example AX means we are referring to the extended 16bit “A” register
- Upper and lower half can be accessed separately,
 - For example
 - AH is for upper 8 bits (A high byte)
 - AL for lower 8 bits (A low byte)
 - AX is used to access whole 16 bits
 - Any changes in AH or AL are reflected in AX.

AH	AL	(AX)
BH	BL	(BX)
CH	CL	(CX)
DH	DL	(DX)

General Registers (AX, BX, CX, and DX)

- AX, A for accumulator, is the accumulator register as it is favored by CPU for arithmetic operations. Provides slightly more efficiency.
- BX, B for base, register can hold the address or procedure or variable. BS can also perform arithmetic or data movement.
- CX, C for counter, acts as counter for repeating or loop instructions. These instructions automatically repeat and decrement CX.
- DX (D for data), has special role in multiply and divide operations, When multiplying, for example, DX holds the high 16 bits of the product.

Index Registers (SI and DI)

- SI and DI stand for source index and destination index respectively
- Hold address of data and used in memory access.
- For flexibility, Intel allows many mathematical and logical operations on these registers as well like the general registers.
- The source and destination are named because of their implied functionality as the source or the destination in a special class of instructions called the string instructions.
- SI and DI are 16bit and cannot be used as 8bit register pairs like AX, BX, CX, and DX.

Instruction Pointer (IP)

- This is the special register containing the address of the next instruction to be executed.
- No mathematics or memory access can be done through this register.
- It is out of our direct control and is automatically used.
- Playing with it is dangerous and needs special care.
- Program control instructions change the IP register

Stack Pointer (SP) and Base Pointer (BP)

- Stack Pointer (SP) is a memory pointer and is used indirectly by a set of instructions. This register will be explored in the discussion of the system stack.
- Base Pointer (BP) It is also a memory pointer containing the address in a special area of memory called the stack and will be explored alongside SP in the discussion of the stack

Flags Register

- The flags register as previously discussed is not meaningful as a unit rather it is bit wise significant and accordingly each bit is named separately. The bits not named are unused. The Intel FLAGS register has its bits organized as follows:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
				O	D	I	T	S	Z		A		P		C

- The individual flags are explained in the next table.
- Empty positions in figure are undefined.

C	Carry	When two 16bit numbers are added the answer can be 17 bits long or when two 8bit numbers are added the answer can be 9 bits long. This extra bit that won't fit in the target register is placed in the carry flag where it can be used and tested.
P	Parity	Parity is the number of "one" bits in a binary number. Parity is either odd or even. This information is normally used in communications to verify the integrity of data sent from the sender to the receiver.
A	Auxiliary Carry	A number in base 16 is called a hex number and can be represented by 4 bits. The collection of 4 bits is called a nibble. During addition or subtraction if a carry goes from one nibble to the next this flag is set. Carry flag is for the carry from the whole addition while auxiliary carry is the carry from the first nibble to the second.
Z	Zero Flag	The Zero flag is set if the last mathematical or logical instruction has produced a zero in its destination.

S	Sign Flag	A signed number is represented in its two's complement form in the computer. The most significant bit (MSB) of a negative number in this representation is 1 and for a positive number it is zero. The sign bit of the last mathematical or logical operation's destination is copied into the sign flag.
T	Trap Flag	The trap flag has a special role in debugging which will be discussed later.
I	Interrupt Flag	It tells whether the processor can be interrupted from outside or not. Sometimes the programmer doesn't want a particular task to be interrupted so the Interrupt flag can be zeroed for this time. The programmer rather than the processor sets this flag since the programmer knows when interruption is okay and when it is not. Interruption can be disabled or enabled by making this bit zero or one, respectively, using special instructions.
D	Direction Flag	Specifically related to string instructions, this flag tells whether the current operation has to be done from bottom to top of the block (D=0) or from top to bottom of the block (D=1).
O	Overflow Flag	The overflow flag is set during signed arithmetic, e.g. addition or subtraction, when the sign of the destination changes unexpectedly. The actual process sets the overflow flag whenever the carry into the MSB is different from the carry out of the MSB

Segment Registers (CS, DS, SS, and ES)

- The code segment register, data segment register, stack segment register, and the extra segment register are special registers related to the Intel segmented memory model and will be discussed later.

Bus and Memory

- 8088 has 8-bit data bus
- It has 20 bit address bus
- It can access 1MB of memory with this 20 bit address bus
 - $2^{20} = 1\text{MB}$

First Program in Assembly language

- The first program that we will write will only add three numbers.
- The addition operation is performed on data present in registers.
- As addition requires two operands, both the operands should be present in registers (or at least one)
- The next slide shows the code with comments.

Example 1.1

```
001 ; a program to add three numbers using registers
002 [org 0x0100]
003     mov ax, 5                ; load first number in ax
004     mov bx, 10               ; load second number in bx
005     add ax, bx               ; accumulate sum in ax
006     mov bx, 15               ; load third number in bx
007     add ax, bx               ; accumulate sum in ax
008
009     mov ax, 0x4c00            ; terminate program
010     int 0x21
```

- Things to note:
 - Abbreviations, for example move is written as mov in assembly operation
 - Code will always start with Line 2 and
 - Code will always ends with Line 9 and 10 to terminate program safely.
 - Instructions are written in following format
 - operation destination, source
 - Code is very much like English language for example
 - mov ax,5 means move 5 to ax

Assemble line and Debug code

- NASM (Netwide Assembler) will be used to assemble the code and create executable file with .com extension
- AFD (A fullscreen debugger) will be used to debug the code.
- DOSBox will be used to emulate 8088 processor.
- Installation of these tools will be discussed in lab.
- Save the code given in previous slide with .asm extension.
- Following command will be used to assemble the code

```
nasm ex01.asm -o ex01.com -l ex01.lst
```

Listing File

- The listing file created has a lot of information. Ans is annotated as follow

	Offset of instruction in hex	OPCODE in hex and Operand in little-endian hex	Assemble code
1			
2			
3	00000000	B80500	[org 0x0100] mov ax, 5
4	00000003	BB0A00	mov bx, 10
5	00000006	01D8	add ax, bx
6	00000008	BB0F00	mov bx, 15
7	0000000B	01D8	add ax, bx
8			
9	0000000D	B8004C	mov ax, 0x4c00
10	00000010	CD21	int 0x21

Size of last instruction (line 10) is 2 bytes so total size of .com file is 16+2=18bytes

Debug

- Following command will be used to debug the file
`afd ex01.com`
- The debugger shows the values of registers, flags, stack, our code, and one or two areas of the system memory as data.
- Debugger allows us to step our program one instruction at a time and observe its effect on the registers and program data.

• Things to note:

- Registers
- Memory (m1 and m2 view)
- Offset of 1st instruction
- IP
- Hit F1 to execute instruction
- See the change in registers/memory

DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: AFD

Register	Value	Register	Value	Register	Value	Register	Value	Stack	Flags
AX	0000	SI	0000	CS	19F5	IP	0100	+0	0000
BX	0000	DI	0000	DS	19F5			+2	20CD
CX	0012	BP	0000	ES	19F5	HS	19F5	+4	9FFF
DX	0000	SP	FFFE	SS	19F5	FS	19F5	+6	EA00

CMD >

Offset	Instruction	Operation	Target
0100	B80500	MOV	AX, 0005
0103	BB0C00	MOV	BX, 000C
0106	01D8	ADD	AX, BX
0108	BB0E00	MOV	BX, 000E
010B	29D8	SUB	AX, BX
010D	B804C	MOV	AX, 4C00
0110	CD21	INT	21
0112	0000	ADD	[BX+SI], AL

Address	0	1	2	3	4	5	6	7
DS:0000	CD	20	FF	9F	00	EA	F0	FE
DS:0008	AD	DE	1B	05	C5	06	00	00
DS:0010	18	01	10	01	18	01	92	01
DS:0018	01	01	01	00	02	FF	FF	FF
DS:0020	FF	FF	FF	FF	FF	FF	FF	FF
DS:0028	FF	FF	FF	FF	EB	19	C0	11
DS:0030	A2	01	14	00	18	00	F5	19
DS:0038	FF	FF	FF	FF	00	00	00	00
DS:0040	05	00	00	00	00	00	00	00
DS:0048	00	00	00	00	00	00	00	00

1 Step 2 ProcStep 3 Retrieve 4 Help ON 5 BRK Menu 6 7 up 8 dn 9 le 10 ri